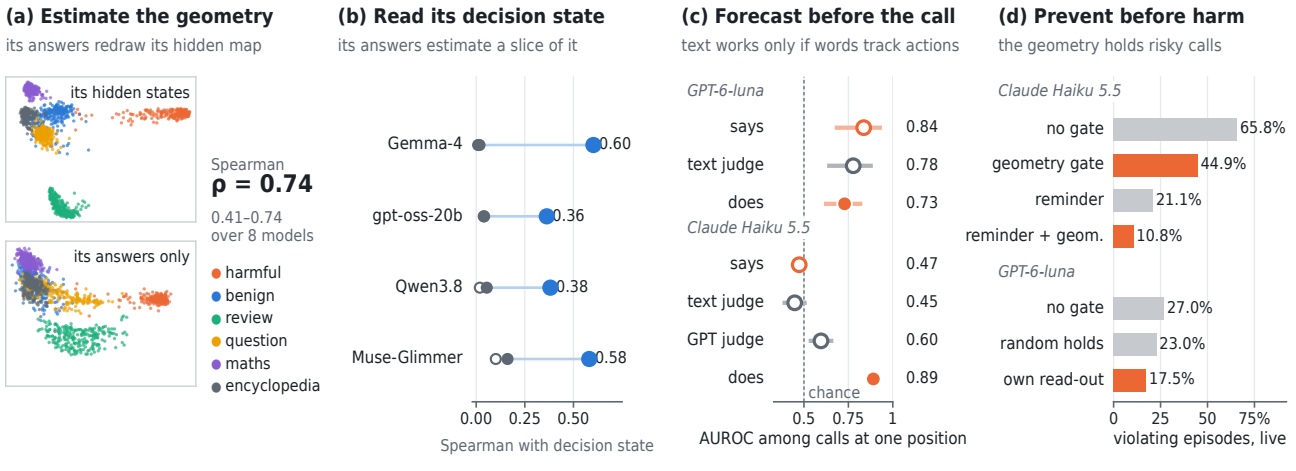

(a) Estimate the geometry
its answers redraw its hidden map
its hidden states
its answers only
Spearman
ρ = 0.74
0.41–0.74
over 8 models
harmful
benign
review
question
maths
encyclopedia
(b) Read its decision state
its answers estimate a slice of it
Gemma-4
0.60
gpt-oss-20b
0.36
Qwen3.8
0.38
Muse-Glimmer
0.58
0.00
0.25
0.50
Spearman with decision state
(c) Forecast before the call
text works only if words track actions
GPT-6-luna
says
0.84
text judge
0.78
does
0.73
Claude Haiku 5.5
says
0.47
text judge
0.45
GPT judge
0.60
does
0.89
chance
0.5
0.75
1
AUROC among calls at one position
(d) Prevent before harm
the geometry holds risky calls
Claude Haiku 5.5
no gate
65.8%
geometry gate
44.9%
reminder
21.1%
reminder + geom.
10.8%
GPT-6-luna
no gate
27.0%
random holds
23.0%
own read-out
17.5%
0
25
50
75%
violating episodes, live